A multiplex genome editing pipeline for rapid combinatorial trait engineering

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Abstract

Current de-extinction efforts center on editing the genomes of extant species to express key traits that evolved in closely related extinct species. These complex projects require advances in genetic engineering pipelines coupled with animal model production as an approach to test hypotheses about genotype and phenotype associations. Here, we establish a multiplex genome engineering pipeline capable of introducing up to eleven genomic modifications across seven different genes simultaneously in viable founder mice. Our workflows achieved high editing efficiencies spanning three genome-editing modalities: Cas9 knockout, Cas9-mediated homology-dependent repair (HDR), and cytosine base editing and included editing of zygotes and embryonic stem cells. As a proof of concept, we produced six mouse models with modifications in genes involved in hair development and lipid metabolism, and the resulting mice displayed predicted hair phenotypes including curly, textured coats, and gold hair. This study advances methods of rapid establishment of complex genetic models, with a wide array of applications.

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